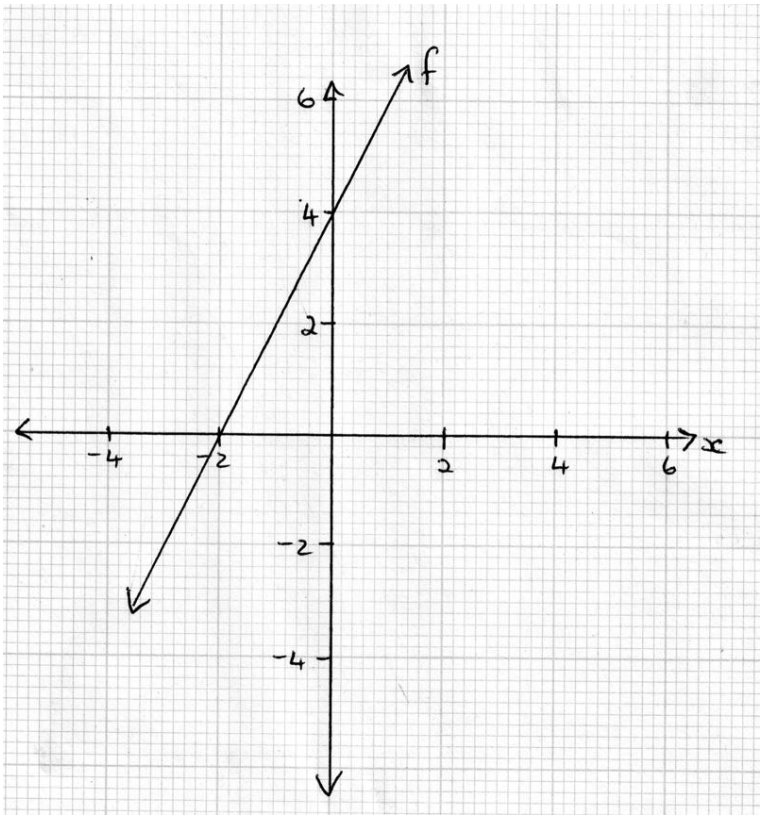
Graph Applications worksheet

EXERCISE A

1. On the set of axes below, the graph of $y = mx + c$ has been drawn.



1.1 Determine the equation of *f* :

1.2. **Use the graph** to answer the following questions:

Find the value of *y* if :

1.2.1 $x = 1$ _____ 1.2.2 $x = -2$ _____

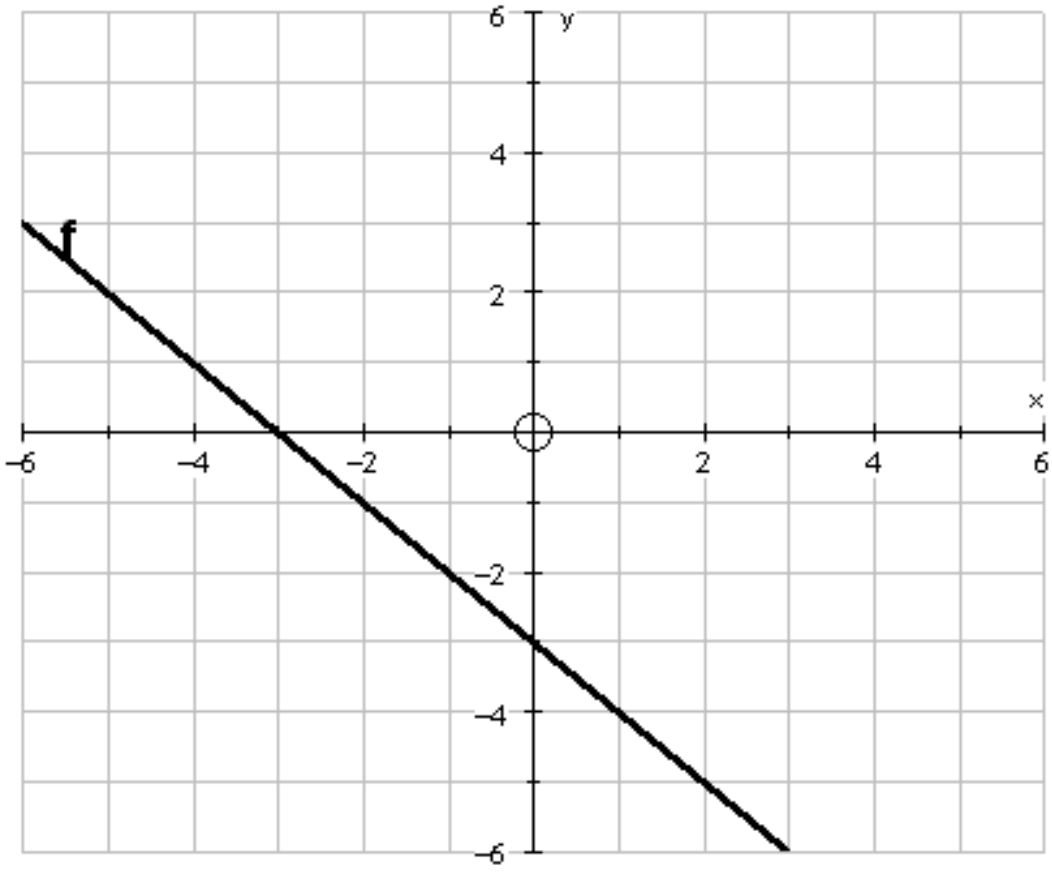
For what value(s) of *x* is:

1.2.3 $y = -2$ _____

1.2.4 $y = 4$ _____

1.3. On the same system of axes, draw the graph of $y = -2x - 2$

2. On the set of axes below, the graph of $y = mx + c$ has been drawn.



2.1 Determine the equation of the graph, *f*:

2.2 **Use the graph** to answer the following questions:

2.2.1 Find the value of *y* if $x = -4$ _____

2.2.3 For what value of x is $y = -2$ _____

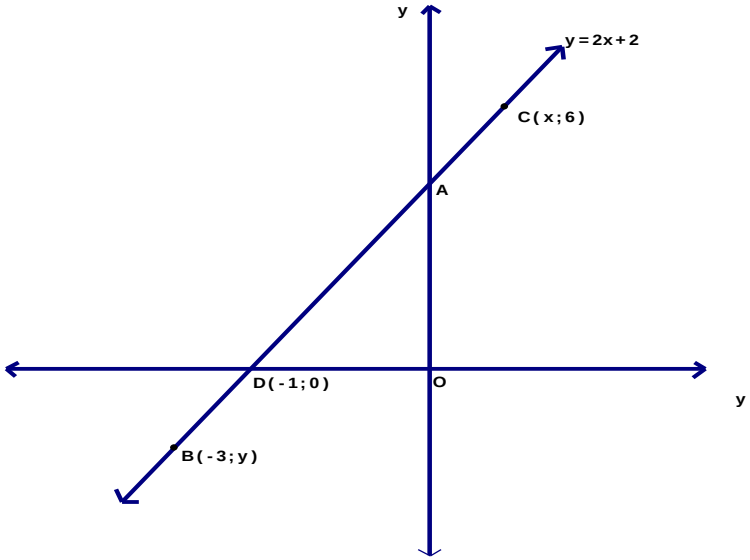
2.3 On the same system of axes, draw the graph of $y = 3x$

Use your graph to answer the following questions:

2.3.1 Find the value of *y* if $x = 1$ _____

2.3.2 For what value of x is $y = 4$ _____

3. On the set of axes below, the graph of $y = 2x + 2$ has been drawn.



3.1 Write down the coordinates of A

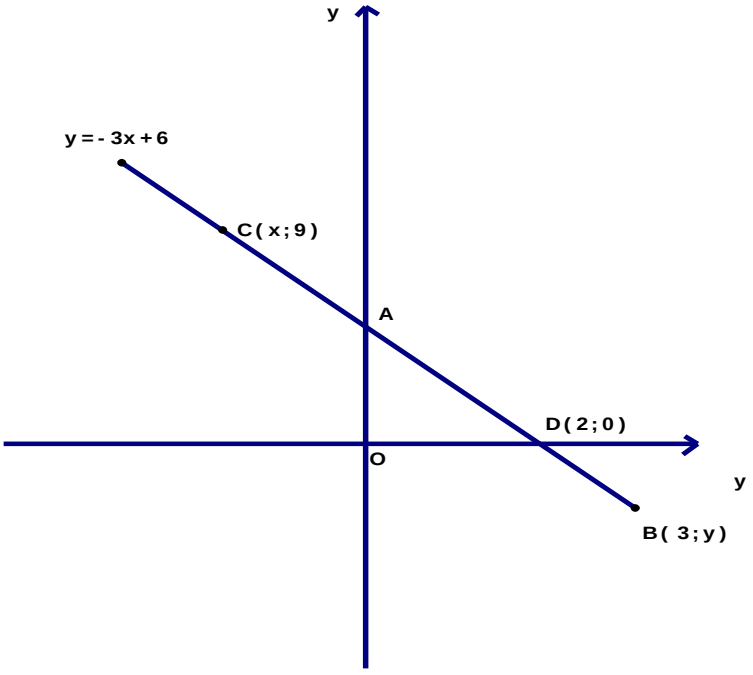
3.2 Determine the *y* coordinate of B.

3.3 Determine the *x* coordinate of C

3.4 Determine the length of AD

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4. On the set of axes below, the graph of $y = -3x + 6$ has been drawn.



4.1. Write down the coordinates of A

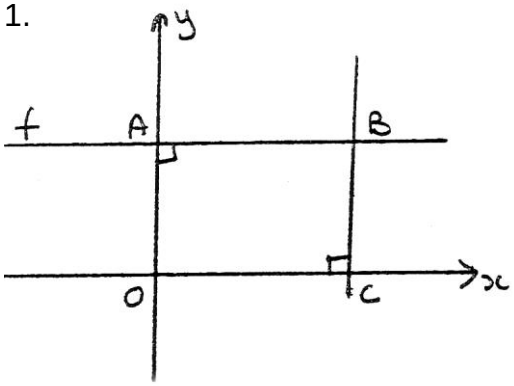
4.2 Determine the *y* coordinate of B.

4.3 Determine the *x* coordinate of C

4.4 Determine the length of AD

EXERCISE B

1.



Equation of AB is $y = 4$

Equation of BC is $x = 3$

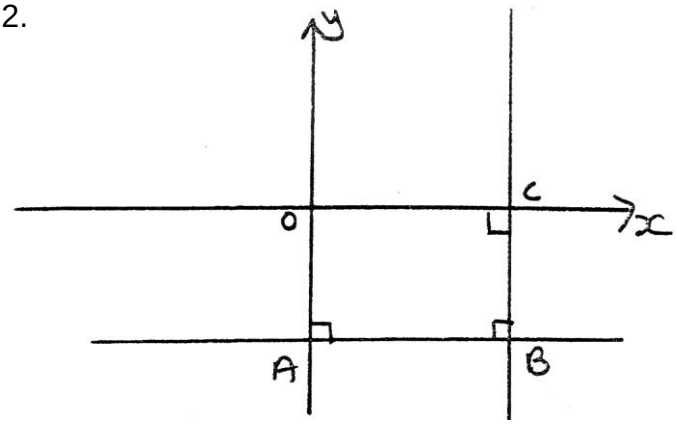
a. the length of AO

b. the length of OC

c. the co-ordinates of B

d. the length of OB.

2.



Equation of AB is $y = -9$

Equation of BC is $x = 12$

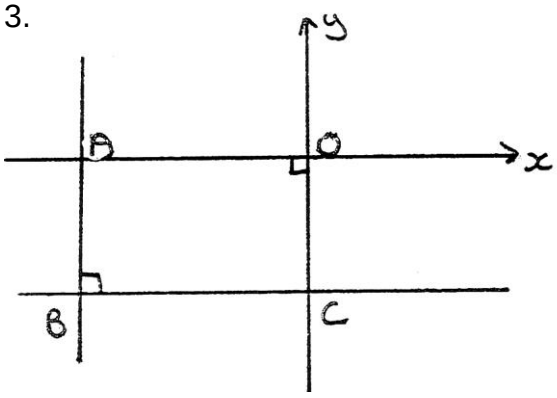
a. the length of AO

b. the length of OC

c. the co-ordinates of B

d. the length of OB.

3.



B is $(-5 ; -12)$

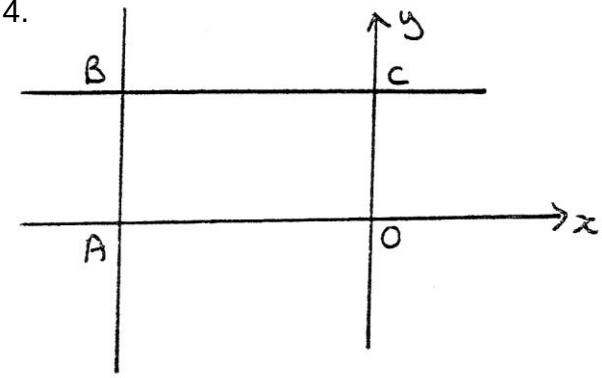
Find:

a. Equation of AB

b. Equation of BC

c. the length of BO.

4.



B is $(-8 ; 15)$

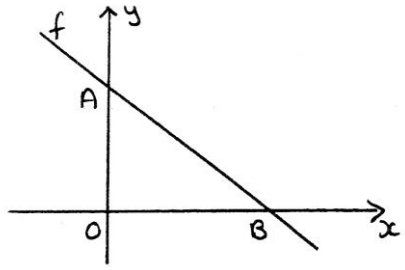
Find:

a. Equation of AB

b. Equation of BC

c. the length of BO.

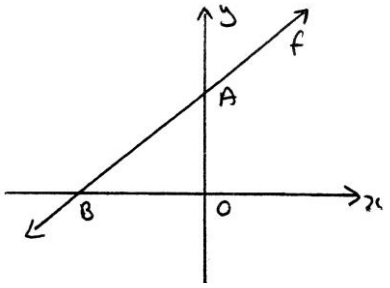
EXERCISE C

1. 

$y = -2x + 4$

Find: a. Length of AO

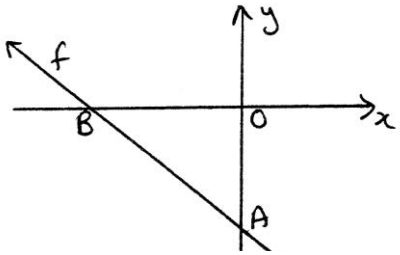
b. Length of OB.

2. 

$y = x + 3$

Find: a. length of AO

b. length of OB.

3. 

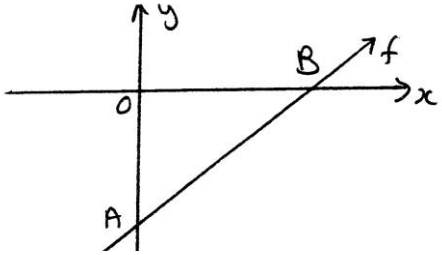
Given: $y = -x - 2$

Find: a. length of OA

b. length of OB

c. length of AB

(Leave answer in surd form.)

4. 

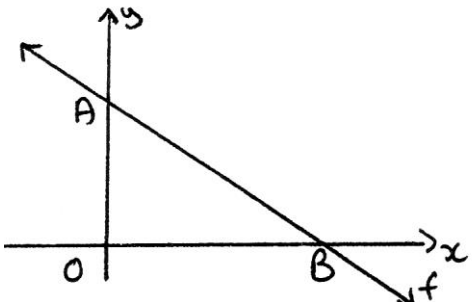
Given: $y = \frac{1}{2}x - 2$

Find: a. length of OA

b. length of OB

c. length of AB

(Leave answer in surd form.)

5. 

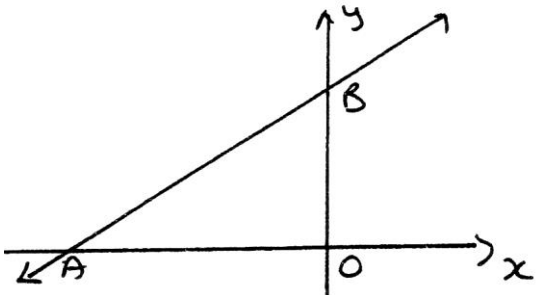
Given: $y = -\frac{1}{5}x + 1$

Find: a. length of OA

b. length of OB

c. length of AB

(Leave answer in surd form.)

6. 

Given: $y = \frac{2}{3}x + 2$

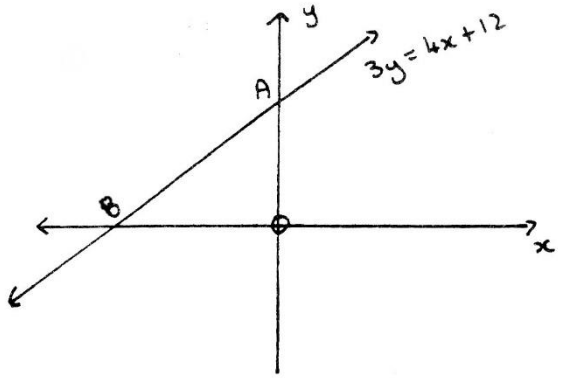
Find: a. length of OB

b. length of OA

c. length of AB

(Leave answer in surd form.)

7.



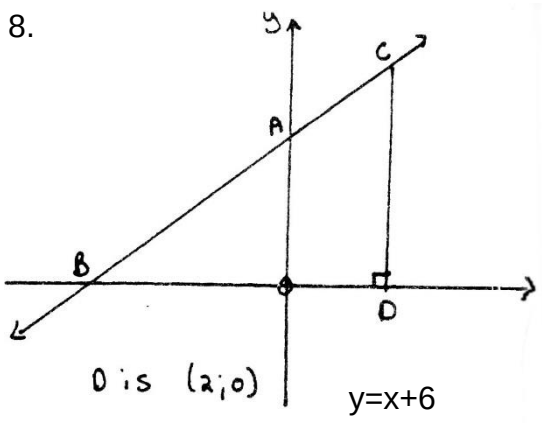
Find:

a. OA

b. OB

c. AB

8.



Find:

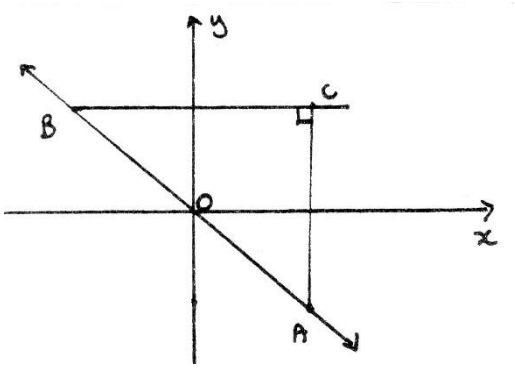
a. CD

b. BD

c. OA

d. BC

9.



The equation of AB is $y = -2x$

and C is $(1 ; 2)$.

Find: (Leave answer in surd form.)

a. AC

b. BC

c. AB